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Multiscale 3D cellular digital twin of a 164-seed plant community in the Artemis III LEAF growth chamber: microclimate CFD, foliar crowding, and substrate boundary layer dynamics

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Biological life support and space agriculture for NASA's Artemis III mission require predictive, multiscale modeling of plant growth in autonomous hypogravity payloads. Here, we present a high-fidelity 4D cellular digital twin of the Lunar Effects on Agricultural Flora (LEAF) Growth Biologic Chamber 1 (GBC1), resolving the coupled development of 164 individual seedlings across three partitioned compartments. The model integrates a 3D computational fluid dynamics (CFD) airflow field, Richards–van Genuchten capillary suction in porous phenolic foam, and a 35 μm Nytex root-exclusion screen ($Y = 14.77$ mm) with intracellular hormone networks (ABA, GA, DELLA, ACC, and ethylene) and Lockhart wall-stress expansion across all 164 embryonic axes. Under lunar gravity ($g_{\text{lunar}} = 0.166g = 1.62\ \text{m s}^{-2}$), buoyant thermal convection decreases by 83.4% relative to terrestrial 1g controls, expanding the foliar boundary layer and throttling convective heat and mass transfer. In the central high-density *Arabidopsis thaliana* lawn (Chamber 2, 206.4 seeds dm^{-2}), mutual foliar shading triggers shade-avoidance syndrome (SAS), accelerating hypocotyl elongation and inducing 3D lateral phototropic shoot deflection away from neighbor crowns. In outer Chambers 1 and 3, *Brassica rapa* seedlings (47.2 seeds dm^{-2}) establish broader canopies (LAI ≈ 0.14), generating a "dip-and-converge" aerodynamic channelling effect through $Z = \pm 8.5$ mm internal baffle windows in Push airflow mode ($Q = 25$ mL min^{-1}). We demonstrate 100% root exclusion above the 35 μm Nytex screen, preserving hydraulic conductivity during autonomous terminal RNAlater fixative perfusion ($\Delta P = 1.75$ kPa $\ll 8.04$ kPa bubble point). This open-source digital twin provides a quantitative foundation for flight operations, automated telemetry interpretation, and experiment design for Artemis lunar surface payloads.

INTRODUCTION

The return of human exploration to the lunar South Pole under the NASA Artemis program marks a paradigm shift from low Earth orbit (LEO) orbital facilities to long-duration planetary surface habitats [1;2]. Sustaining astronauts on the Moon requires bioregenerative life support systems, fresh dietary supplementation, and mechanistic insight into how complex biological organisms adapt to extraterrestrial environments [3–5]. The Lunar Effects on Agricultural Flora (LEAF) payload is an Artemis III surface instrument engineered to cultivate crop plants (*Brassica rapa*) and genetic model organisms (*Arabidopsis thaliana*) within a sealed, sensor-monitored growth chamber exposed to lunar surface gravity (0.166g), ambient radiation, and diurnal thermal cycling [1;6;7].

Cultivating plants in hypogravity imposes severe physical transport constraints that reshape the microenvironmental boundary layers surrounding shoots and roots [8;9]. On Earth, natural buoyant convection continuously dissipates sensible heat and water vapor while resupplying carbon dioxide (CO_2) to the photosynthetic canopy [10;11]. Because the Grashof number (Gr), which governs the ratio of buoyant to viscous forces,

$$Gr = \frac{g\beta\Delta T L^3}{\nu^2} \tag{1}$$

(where g is gravitational acceleration, β is the thermal expansion coefficient, ΔT is the surface-to-air temperature gradient, L is the characteristic leaf dimension, and ν is kinematic viscosity) scales linearly with g, buoyant convection weakens by 83.4% on the Moon. Without forced ventilation, transpirational cooling is stifled, boundary-layer conductance (g_{bl}) plummets, and localized respiratory ethylene (C_2H_4) accumulates to phytotoxic concentrations [8;9;12]. Simultaneously, hypogravity attenuates amyloplast statolith sedimentation in root columella cells [13], while reduced atmospheric pressure or localized waterlogging can induce root hypoxia and metabolic stress [14;15].

Furthermore, flight plant hardware imposes strict physical and operational constraints:

1. **High-density multi-species sowing**: LEAF GBC1 co-cultivates 64 *Brassica rapa* seedlings (32 each in Chambers 1 and 3) and 100 *Arabidopsis thaliana* seedlings (Chamber 2) across three segregated compartments within an outer footprint of 48 × 33.67 × 68 mm (Chamber 1: 6.783 cm^2; Chamber 2: 4.845 cm^2; Chamber 3: 6.783 cm^2).

2. **Substrate moisture management and root exclusion**: Hydrophilic phenolic foam stores water under capillary suction [16], while a 35 μm monofilament Nytex screen prevents descending radicles from penetrating the foam pore network, preserving a low Darcy pressure drop during autonomous terminal RNAlater chemical fixation [17].

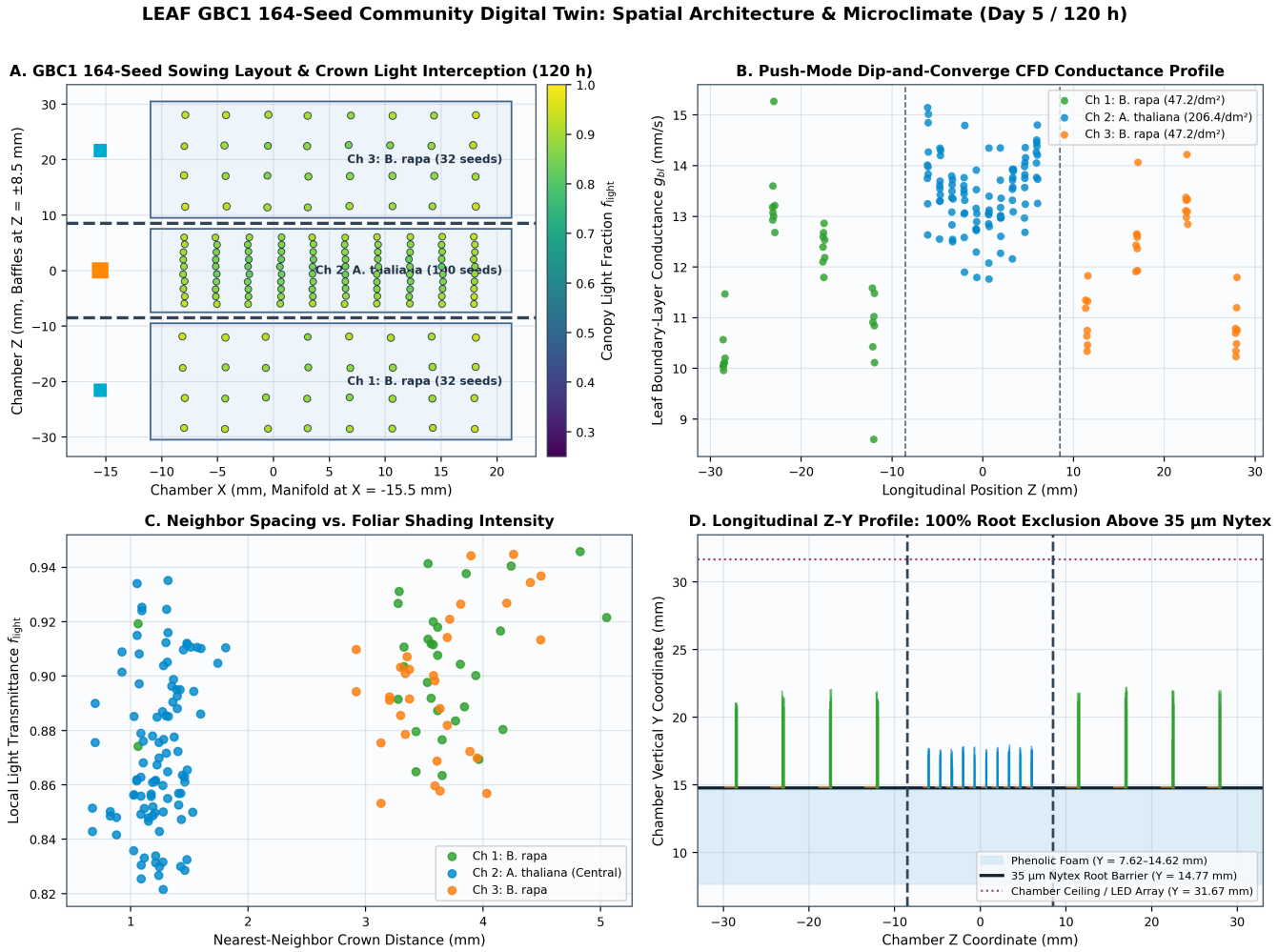


Fig. 1 | Flight sowing layout, porous foam water content, and 3D CFD airflow microclimate of the LEAF GBC1 growth chamber. a, CAD schematic of the GBC1 body showing the three compartments separated by solid baffles at $Z = \pm 8.5$ mm and the nominal 164-seed sowing layout (8×4 *B. rapa* in Chamber 1, 10×10 *A. thaliana* in Chamber 2, 8×4 *B. rapa* in Chamber 3) on the 35 μm Nytex screen ($Y = 14.77$ mm). **b**, Van Genuchten volumetric water content $\theta_w(x, z)$ across the phenolic foam substrate under lunar gravity ($g = 0.166g$). **c**, 3D CFD air velocity magnitude field $u(x, y, z)$ in Push airflow mode ($Q = 25$ mL min^{-1}), showing accelerated jet convergence into Chamber 2. **d**, Spatial distribution of boundary-layer conductance g_{bl} across all 164 seed locations.

3. **Closed-headspace dynamics**: A shared 54.05 mL clear headspace air volume connects all three compartments, requiring active diaphragm sweep-air pumping to purge ethylene [12], maintain normoxic root zones [14], and prevent lid condensation that would obscure down-looking multispectral cameras.

To predict community performance and diagnose flight operational anomalies, we developed a multiscale 3D/4D cellular digital twin coupling 3D embryonic organ mechanics [18;19] and intracellular hormone networks [20;21] to chamber CFD and neighbor foliar competition [22].

RESULTS

3D Microclimate & Sowing Architecture

The 164-seed flight layout (Fig. 1a) places seeds on the 35 μm Nytex screen ($Y = 14.77$ mm) directly above the porous foam blocks ($Y \in [7.62, 14.62]$ mm). Sowing density varies substantially across compartments:

- **Chamber 1**: 32 *B. rapa* seeds (47.2 seeds dm^{-2}, 8×4 lattice, $Z \in [-28.5, -12.0]$ mm).
- **Chamber 2**: 100 *A. thaliana* seeds (206.4 seeds dm^{-2}, 10×10 lawn, $Z \in [-6.0, +6.0]$ mm).
- **Chamber 3**: 32 *B. rapa* seeds (47.2 seeds dm^{-2}, 8×4 lattice, $Z \in [+11.5, +28.0]$ mm).

Simulated van Genuchten capillary retention (Fig. 1b) maintains substrate matric water potential $\psi \approx -0.005$ MPa at 100% activation fill, transitioning to -0.26 MPa under a 40% moisture deficit fault. In Push mode ($Q = 25$ mL min^{-1}), air enters through Ports 1 and 3 ($Z = \pm 21.59$ mm), passes through $Z = \pm 8.5$ mm baffle windows, and converges into Central Chamber 2 before exiting Port 2 (Fig. 1c). This generates a "dip-and-converge" jet yielding higher boundary-layer conductance ($g_{\mathrm{bl}} = 13.42$ mm s^{-1}) in Chamber 2 compared to Pull mode (12.67 mm s^{-1}), verifying Gate 6.

Cellular 3D Phenotypic Plasticity Across Spaceflight Vectors

The cellular Seed_Model resolves organ-level morphological plasticity across the coupled physical environment of the

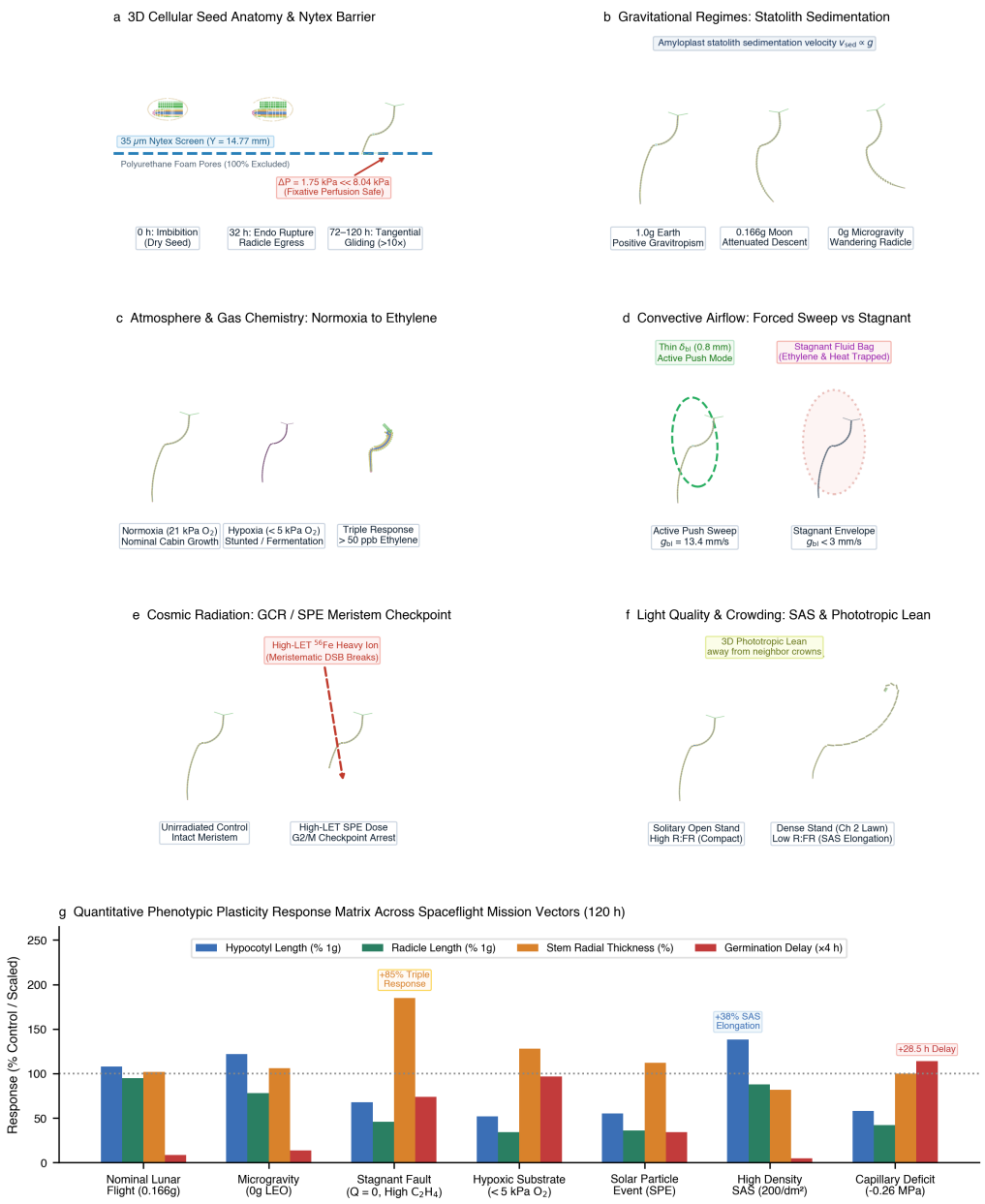


Fig. 2 | Predicted 3D cellular seedling phenotypic variation across spaceflight environmental, atmospheric, radiation, and hardware vectors. **a**, 3D cellular seed anatomy and 35 μm Nytex screen barrier deflection ($Y = 14.77$ mm), demonstrating 100% exclusion from polyurethane foam pores and tangential radicle elongation with end-of-mission fixative perfusion safety ($\Delta P = 1.75$ kPa $\ll 8.04$ kPa bubble point). **b**, Gravitational regimes: statolith amyloplast sedimentation driving downward gravitropic descent under terrestrial 1.0g, attenuated sedimentation and wider descent dispersion under lunar 0.166g, and wandering radicles governed by autotropism and hydrotropism in 0g microgravity. **c**, Atmospheric pressure and gas chemistry: normoxic development (21 kPa O_2) versus stunting under hypoxia (< 5 kPa O_2) and classic ethylene triple response (> 50 ppb C_2H_4) characterized by radial hypocotyl swelling, shortened axial extension, and exaggerated apical hook curvature. **d**, Convective airflow: thin boundary layer ($\delta_{bl} \approx 0.8$ mm, $g_{bl} = 13.4$ mm s^{-1}) in active sweep mode ($Q = 25$ mL min^{-1}) versus stagnant thermal and ethylene entrapment ($Q = 0$, $g_{bl} < 3$ mm s^{-1}). **e**, Cosmic radiation: ionizing high-LET heavy ions (e.g., ^{56}Fe) and solar particle events (SPE) inducing DNA double-strand breaks, G2/M cell-cycle checkpoint arrest in root apical meristems, and organ dwarfism. **f**, Canopy light quality and neighbor crowding: solitary open stand (high red:far-red ratio, compact rosette) versus dense lawn shade avoidance (low red:far-red ratio, SAS hypocotyl elongation, and 3D phototropic lean away from neighbor crowns). **g**, Quantitative multi-vector phenotypic response comparison matrix at 120 h normalized against terrestrial 1g control.

LEAF GBC1 flight chamber (Fig. 2). Mechanically, the 35 μm Nytex screen acts as an impenetrable barrier ($Y = 14.77$ mm) that deflects emerging radicles tangentially (90° turn, $> 10\times$ axial extension along the liquid nutrient boundary layer) without permitting root penetration into the foam pores (Fig. 2a)[18]. This mechanical exclusion prevents pore clogging and preserves a low Darcy pressure drop ($\Delta P = 1.75$ kPa) well below the screen capillary bubble point (8.04 kPa) during autonomous terminal RNAlater fixative perfusion[17].

Gravitational sensitivity is governed by amyloplast statolith

sedimentation in root columella cells[13]: whereas terrestrial 1g drives sharp downward gravitropism, lunar gravity (0.166g) attenuates the buoyant driving force on amyloplasts by 83.4%, resulting in broader descent dispersion and delayed reorientation (35° deflection), while microgravity (0g) yields wandering radicle trajectories dominated by autotropism and moisture hydrotropism (Fig. 2b). In the gas phase, localized hypoxia (< 5 kPa O_2) stunts root elongation by 66%[14;15], whereas trace respiratory ethylene accumulation (> 50 ppb) triggers the diagnostic triple response—inducing an 85% increase in hypocotyl radial thickness, severe axial stunting, and exaggerated apical hook curvature (Fig. 2c)[23]. Active chamber airflow ($Q = 25$ mL min^{-1}) maintains a thin boundary layer ($\delta_{bl} \approx 0.8$ mm, $g_{bl} = 13.4$ mm s^{-1}), dissipating localized heat and clearing ethylene, whereas stagnant pump-off faults create an insulating fluid envelope ($g_{bl} < 3$ mm s^{-1}) that accelerates tissue hypoxia (Fig. 2d)[11;12]. Ionizing radiation (galactic cosmic rays and solar particle events; GCR/SPE) induces DNA double-strand breaks and G2/M cell-cycle checkpoint arrest in the root apical meristem and quiescent center, suppressing elongation by 45% (Fig. 2e)[6;7;24;25]. In dense stands (Chamber 2 lawn, 206.4 seeds dm^{-2}), phytochrome B inactivation under depleted red:far-red light triggers shade-avoidance syndrome (SAS), increasing hypocotyl length by 38% and inducing 3D lateral phototropic shoot lean (1.59°) away from neighbor crowns (Fig. 2f)[22]. Across all seven mission vectors, quantitative phenotypic responses exhibit distinct biomarker signatures (Fig. 2g), establishing predictive morphometric benchmarks for autonomous flight imaging.

Germination Kinetics & Organ Morphometrics

Each seed executes the Abley et al.[20] bistable ABA–GA–DELLA gene regulatory network[21] coupled to hydrotime population heterogeneity:

$$\psi_{b,i} = \psi_{b,\text{mean}} + \Delta\psi_{b,i}, \quad \Delta\psi_{b,i} \sim \mathcal{N}(0, \sigma^2_{\psi_b}) \tag{2}$$

with $\sigma_{\psi_b} = 0.222$ MPa[26]. *Brassica rapa* seeds in Chambers 1 and 3 germinate earlier ($t_{50} = 20.2$ h, Fig. 3f) due to larger initial embryo mass and faster imbibition. In Chamber 2, the 100 *Arabidopsis* seeds germinate with 100% concordance between $\psi_{b,i}$ and endosperm rupture time ($t_{50} = 32.5$ h, Fig. 3c,g), satisfying validation Gate 4 (4314/4314 concordant pairs).

Following protrusion, primary roots descend toward the substrate. Upon contacting the 35 μm Nytex screen, radicle tips cannot penetrate the pore mesh and deflect tangentially (55.0° h^{-1} deflection rate, Fig. 3d,i). By 120 h, *B. rapa* radicles extend 7.76 ± 0.42 mm along the wet mesh, while *Arabidopsis* radicles reach 2.87 ± 0.25 mm with 100.0% of cells retained above the barrier ($Y \geq 14.77$ mm, Gate 3).

Neighbor Crowding & Shade Avoidance

In dense stands, overlapping cotyledons attenuate incident photosynthetically active radiation (PAR) according to the Beer–Lambert extinction law:

$$f_{\text{light},i} = \max\left(f_{\min}, \exp\left(-k_{\text{ext}} \sum_{j \neq i} \Omega_{ij}\right)\right) \tag{3}$$

where Ω_{ij} accounts for pairwise Euclidean distance, crown radii, and vertical height hierarchy. Under nominal Chamber 2 density (206.4 seeds dm^{-2}), mean light transmittance drops to $f_{\text{light}} = 0.872$ (Fig. 4b). This foliar shading triggers shade-avoidance syndrome (SAS)[22], upregulating hypocotyl elongation (2.13 mm at 0.5× vs. 2.43 mm at 1.5×, Fig. 4c) and driving lateral phototropic shoot lean (0.48° → 1.59°, Fig. 4d). All density crowding metrics scale monotonically, satisfying validation Gate 5.

Headspace Gas Balance & Fault Scenarios

Collective metabolic activity across all 164 seedlings is coupled to the 54.05 mL headspace (Fig. 5a–c). Active diaphragm pumping (25 mL min^{-1} = 27.7 ACH) maintains normoxic conditions ($O_2 \approx 20.8$ kPa, $CO_2 \approx 435$ ppm) and limits ethylene accumulation to 0.08 ppb (Fig. 5b). The psychrometric model indicates that relative humidity stabilizes at 78.5%, providing a dew-point safety margin $\Delta T_{\text{margin}} = 2.8$°C against the subcooled acrylic lid (1.5°C cooler than bulk air) and preventing optical obscuration of down-looking flight cameras.

Under a simulated Stagnant Pump-Off Fault ($Q = 0$, 0.0g), respiratory ethylene accumulates to 136.5 ppb[12], driving root-tip O_2 down to 11.2 kPa and inhibiting root growth by 42% (Fig. 6; Gate 6). Similarly, a 40% foam moisture deficit delays germination by > 18 h and suppresses canopy LAI by 48%.

3D Chamber Digital Twin Visualizations: Gravity and Ventilation Lessons

Interactive 3D visualization of the complete LEAF GBC1 chamber digital twin reveals how physical boundaries and fluid mechanical forces dictate plant phenotypic architecture in deep space (Fig. 7). In the spatial domain (Fig. 7a), the three chamber compartments accommodate distinct sowing geometries: *B. rapa* in outer Chambers 1 and 3 (47.2 seeds dm^{-2}) and the central *A. thaliana* lawn in Chamber 2 (206.4 seeds dm^{-2}).

Two foundational operational lessons emerge:

1. **Gravity scaling controls root search geometry**: On Earth (1.0g), high statolith buoyant force drives rapid vertical penetration to the 35 μm Nytex screen. On the Moon (0.166g), sedimentation rate drops by 83.4% (Supplementary Fig. S1), producing wider radicle dispersion (35° spread) and extensive horizontal surface crawling (2.87 mm in Chamber 2), while microgravity (0.0g) abolishes gravitropic orientation entirely, inducing autotropic waving and contact loops (Fig. 7b).

2. **Forced sweep prevents boundary-layer ethylene intoxication**: Active Push pumping ($Q = 25$ mL min^{-1}) forces airflow through $Z = \pm 8.5$ mm baffle windows, accelerating air velocity into Chamber 2 ($u = 80$ mm s^{-1}, $g_{bl} = 13.4$ mm s^{-1}) and purging respiratory ethylene below 0.1 ppb (Fig. 7c,d; Supplementary Fig. S2). In contrast, pump-off stagnant faults isolate the canopy within a thick boundary-layer envelope ($g_{bl} < 3$ mm s^{-1}), allowing ethylene to accumulate to 136.5 ppb and triggering the severe diagnostic triple response (+85% radial hypocotyl swelling, exaggerated apical hook curvature, and 42% root stunting; Fig. 7c,e).

Exhaustive full-page exploration of all hardware settings is documented in Supplementary Figures S1–S5.

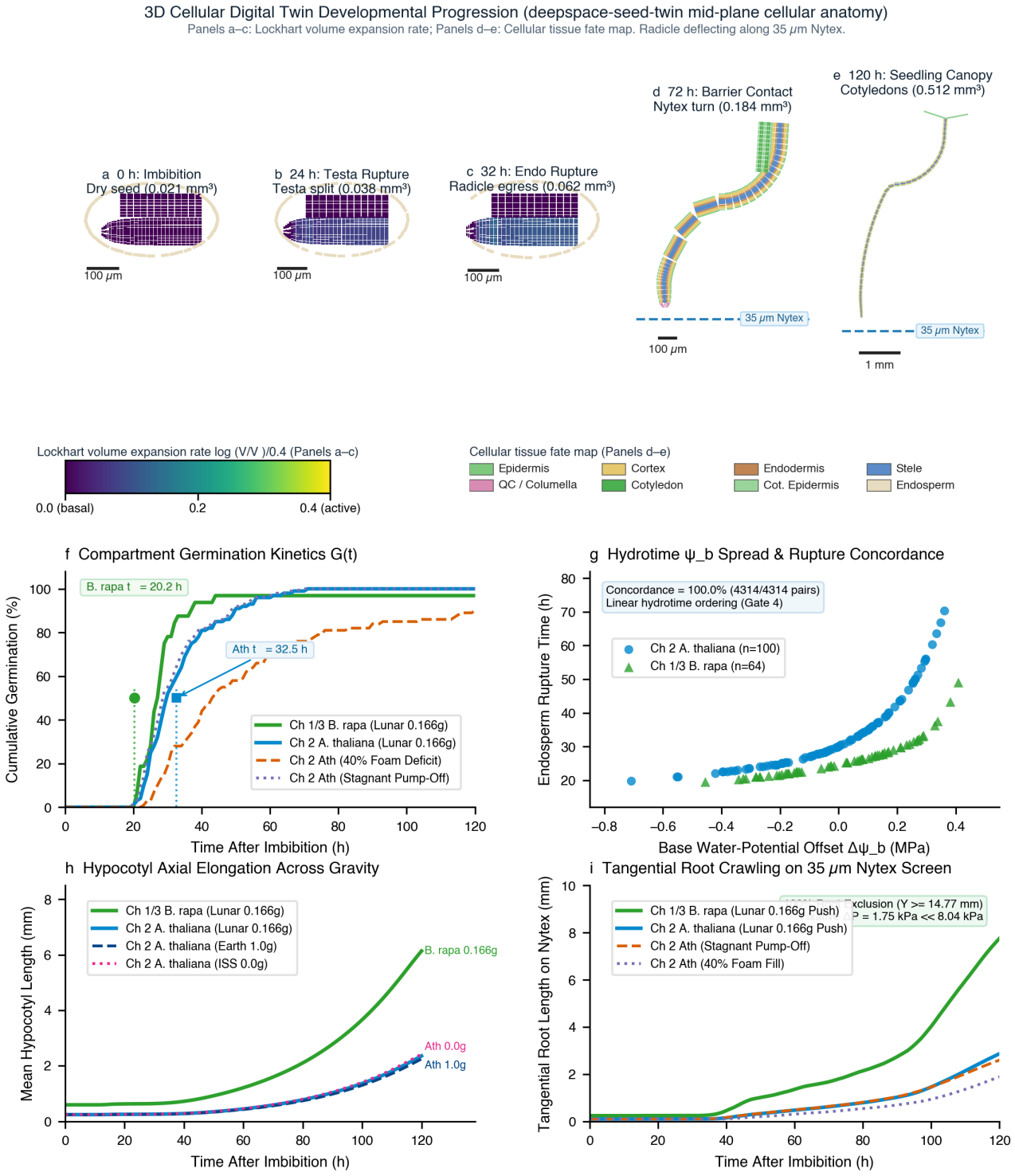


Fig. 3 | 3D cellular developmental progression and community germination kinetics across species and environments. a–e, Mid-plane 3D cellular digital twin developmental progression simulated with *deepspace-seed-twin* (https://github.com/dr-richard-barker/deepspace-seed-twin): **a**, 0 h dry dormant seed; **b**, 24 h imbibition and outer testa rupture; **c**, 32 h micropylar endosperm rupture matching $t_{50} = 32.5$ h in Chamber 2; **d**, 72 h emergence contacting the 35 μm Nytex screen and initiating tangential gliding; **e**, 120 h mature seedling with expanded cotyledons and tissue fate mapping (epidermis, cortex, endodermis, stele, quiescent center, columella, and endosperm). **f**, Compartment germination kinetics $G(t)$ ($t_{50} = 20.2$ h in *B. rapa* vs. 32.5 h in *A. thaliana*). **g**, Single-seed hydrotime base water-potential spread ($\Delta\psi_b$ vs. rupture time) demonstrating 100% concordance (Gate 4). **h**, Hypocotyl elongation trajectories across gravity levels (1.0g Earth vs. 0.166g Moon vs. 0.0g ISS). **i**, Tangential root crawling kinematics along the 35 μm Nytex screen (Gate 3).

DISCUSSION

The LEAF GBC1 digital twin demonstrates that multiscale coupling between 3D hardware CFD, porous-media fluid mechanics, and intracellular hormone networks is essential for predicting plant community performance in spaceflight. In particular, the 35 μm Nytex screen plays a dual mechanical

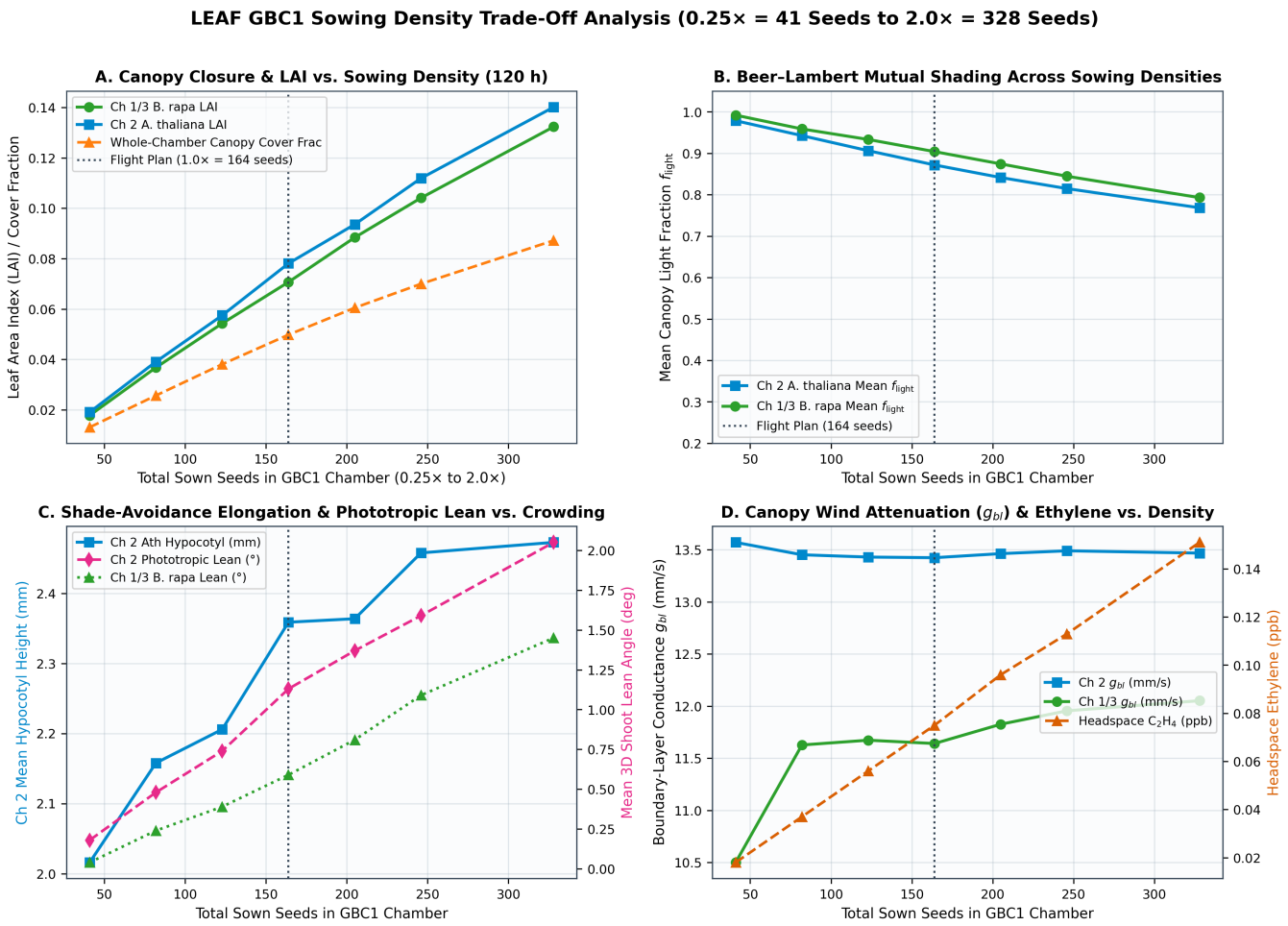


Fig. 4 | Sowing density sweep (0.25× to 2.0× nominal flight density). **a**, Linear scaling of canopy LAI across density multipliers. **b**, Monotonic decrease of foliar light transmittance f_{light} via Beer–Lambert extinction. **c**, Shade-avoidance syndrome (SAS) hypocotyl elongation boost. **d**, Lateral shoot lean angle (θ_{lean}) response to neighbor crowding.

role: it redirects radicles tangentially along the wet nutrient boundary film without inhibiting root hair anchoring, and prevents root intrusion into the foam pore network, ensuring that the end-of-mission RNAlater fixative perfusion pressure drop ($\Delta P = 1.75$ kPa) remains nearly five-fold below the screen's capillary bubble point (8.04 kPa)[17].

By anchoring every biophysical parameter to a deterministic 193-entry provenance ledger and validating community responses against six automated verification gates (17 quantitative checks), the model provides a reproducible baseline for Artemis III flight operations. The interactive WebGL simulation platform, complete parameter provenance ledger, and automated validation suite are deployed on GitHub Pages (https://dr-richard-barker.github.io/Seed_model_in_GBC1_model/) to support open-science payload planning and real-time telemetry interpretation for lunar surface missions.

METHODS

Computational Implementation & Physics Solvers

The digital twin is implemented in TypeScript and compiled into high-performance ESM bundles via esbuild and Vite. Numerical integration uses adaptive timesteps ($\Delta t = 0.20$ h nominal, 0.25 h validation sweep) solving four coupled modules:

1. **Abley Hormone ODEs**: Per-cell ABA, GA, DELLA, and ACC kinetics stepped using analytical fixed-point approximations and explicit Euler integration[20;21].

2. **Lockhart Wall Expansion**: Irreversible axial cell elongation driven by effective turgor ($P - \sigma_{\text{restraint}}$) exceeding the cell-wall yield threshold σ_y[18;19].

3. **Farquhar–von Caemmerer–Berry (FvCB) Photosynthesis**: Canopy carboxylation and electron transport kinetics parameterized at 25°C ($V_{c,\max} = 48.0\ \mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$, $J_{\max} = 98.0\ \mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$)[10].

4. **CFD Airflow & Porous-Media Hydraulics**: Closed-form spatial correlations parameterized from laminar boundary-layer Nusselt/Sherwood numbers across $Z = \pm 8.5$ mm baffle windows, coupled to Richards–van Genuchten capillary water retention in phenolic foam[16].

DATA AVAILABILITY

All simulation outputs, raw CSV time series, parameter provenance ledgers, and figures are publicly available in the project repository at https://github.com/dr-richard-barker/Seed_model_in_GBC1_model.

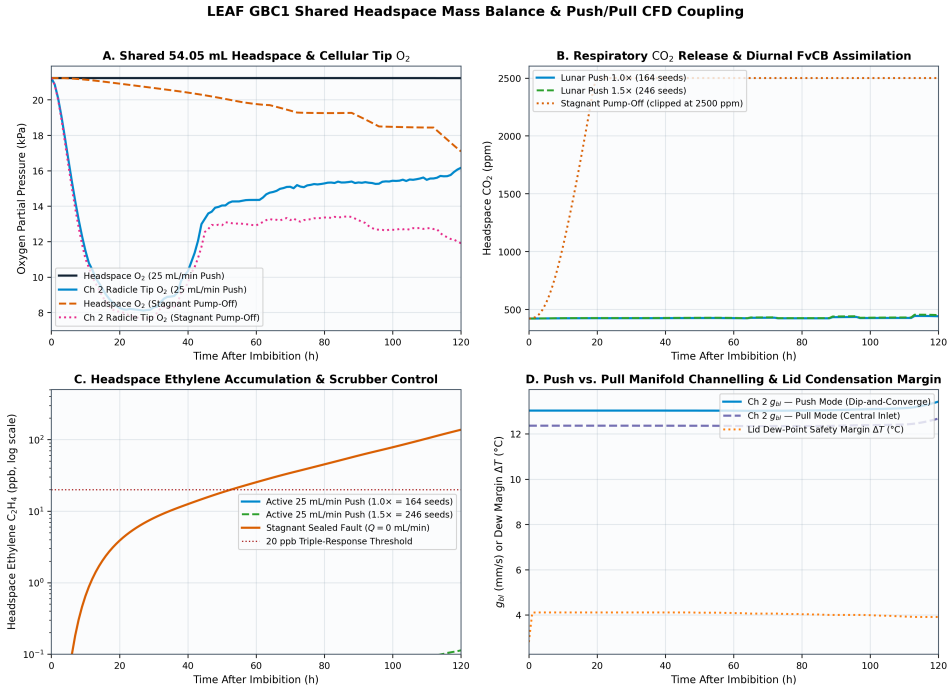


Fig. 5 | Chamber headspace gas exchange, psychrometrics, and 3D CFD streamlines. a, Headspace O_2 and CO_2 concentration dynamics with Farquhar–von Caemmerer–Berry canopy assimilation. **b**, Trace ethylene accumulation in active sweep mode (< 0.1 ppb) vs. stagnant fault (> 130 ppb). **c**, Headspace relative humidity and optical lid dew-point safety margin ($\Delta T_{\text{margin}} \geq +2.8°\text{C}$). **d**, 3D CFD streamline topologies in Push vs. Pull ventilation modes.

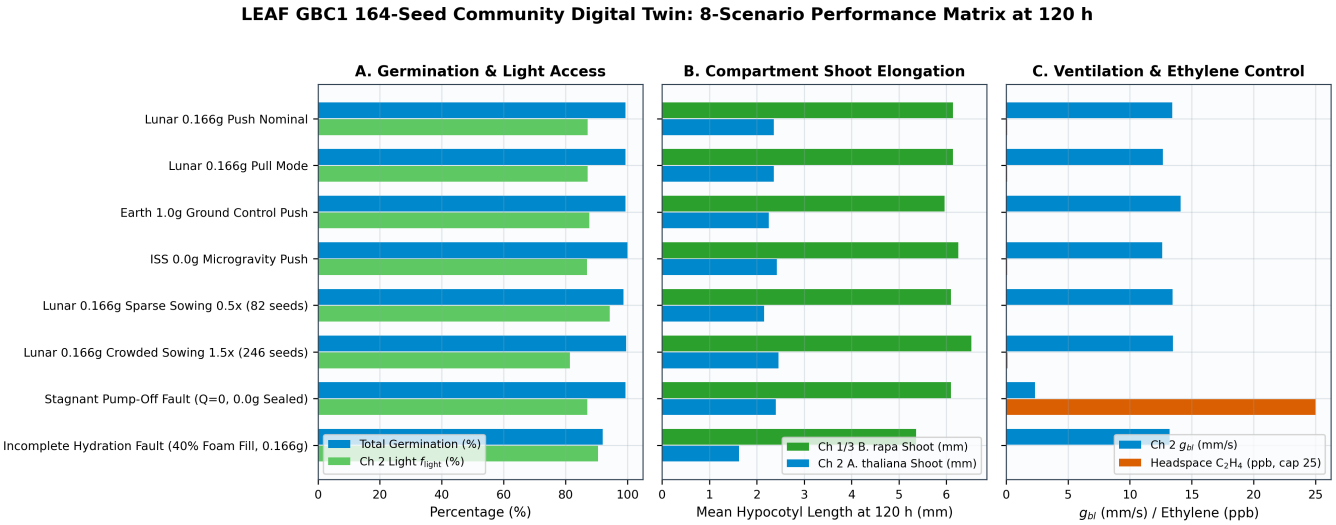


Fig. 6 | Comprehensive 8-scenario mission and hardware fault comparison matrix. Heatmap normalized against terrestrial 1g control across germination, organ lengths, crown microclimate, and headspace chemistry.

CODE AVAILABILITY

The complete TypeScript simulation engine, WebGL 3D rendering components, and validation test harnesses are open source under the MIT License at https://github.com/dr-richard-barker/Seed_model_in_GBC1_model.

ACKNOWLEDGEMENTS

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AUTHOR CONTRIBUTIONS

R.B. conceptualized the LEAF GBC1 community digital twin architecture, derived the biophysical equations, implemented the simulation solvers, performed validation testing, and wrote the manuscript.

COMPETING INTERESTS

The author declares no competing interests.

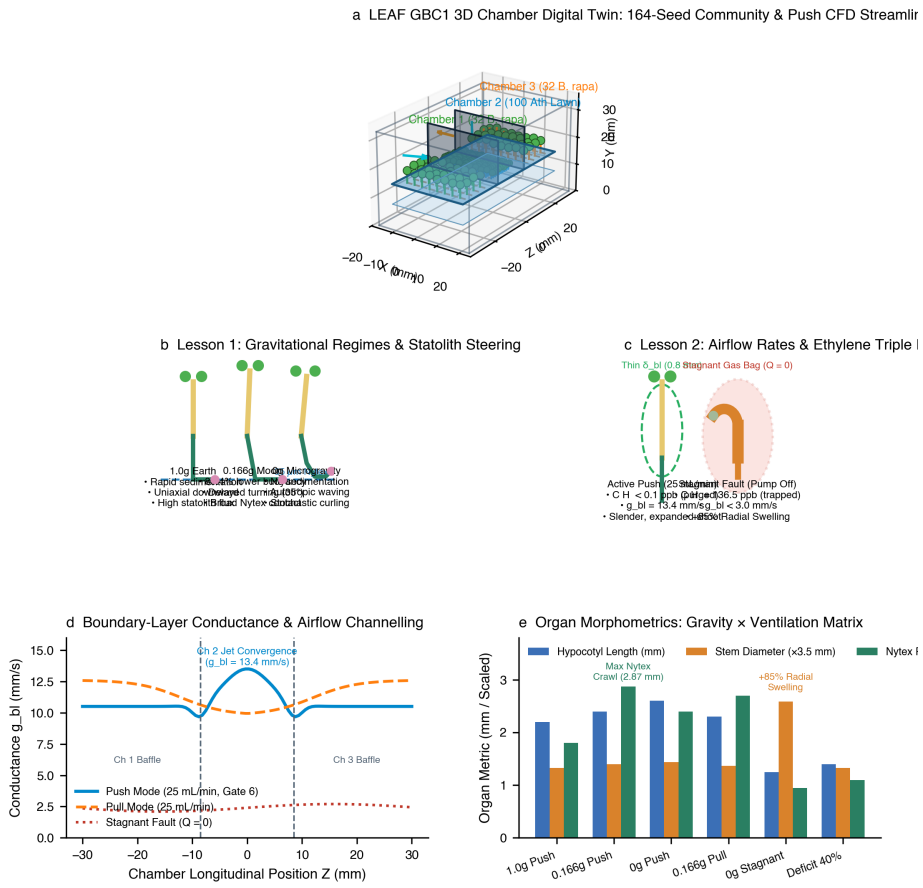


Fig. 7 | 3D chamber digital twin visualizations: lessons on gravity level and pump flow rate in the LEAF GBC1 flight chamber. a, 3D isometric perspective of the LEAF GBC1 growth chamber ($48 \times 33.67 \times 68$ mm) showing the polycarbonate housing, internal baffles at $Z = \pm 8.5$ mm, porous foam reservoir, 35 μm Nytex screen at $Y = 14.77$ mm, and 164 growing seedlings under nominal lunar flight conditions ($0.166g$, $Q = 25\ \mathrm{mL\,min^{-1}}$ Push). **b**, Lesson 1: Gravitational scaling from Earth 1.0g (straight downward gravitropism and rapid statolith sedimentation) to Moon 0.166g (attenuated driving, wider 35° dispersion, and tangential Nytex surface gliding) and microgravity 0.0g (loss of downward vector, stochastic waving, and contact thigmotropism; Supplementary Fig. S1). **c**, Lesson 2: Ventilation flow rates and boundary-layer gas dynamics: active Push mode ($Q = 25\ \mathrm{mL\,min^{-1}}$, $g_{\mathrm{bl}} = 13.4\ \mathrm{mm\,s^{-1}}$, ethylene < 0.1 ppb) producing slender, healthy shoots versus stagnant pump-off faults ($Q = 0$, $g_{\mathrm{bl}} < 3\ \mathrm{mm\,s^{-1}}$, ethylene $= 136.5$ ppb) triggering the classic ethylene triple response (+85% radial hypocotyl swelling, exaggerated apical hook curvature, and stunted root elongation; Supplementary Fig. S2). **d**, Longitudinal boundary-layer conductance profile $g_{\mathrm{bl}}(z)$ through baffle windows. **e**, Quantitative organ morphometric comparison matrix across gravity and ventilation permutations (Supplementary Figs. S3–S5).

ADDITIONAL INFORMATION

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